

Derivatives, Tangents, Extremal Problems, Young–Holder–Minkowski, and Jensen

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§1.1 The derivative from secants to tangents

The derivative begins with the geometry of two nearby points on a curve.

$$(x_0, f(x_0)), \quad (x_0 + \Delta x, f(x_0 + \Delta x)),$$

the secant slope is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

If, as $\Delta x \rightarrow 0$, this quotient tends to a finite limit, then the curve has a definite tangent line at x_0 . The derivative is

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

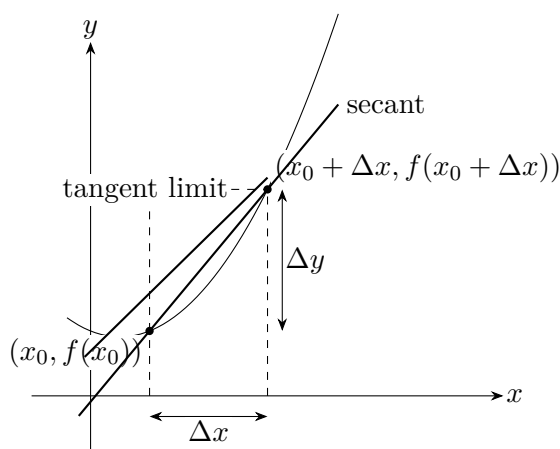
Equivalently,

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

The tangent line at x_0 is therefore

$$y - f(x_0) = f'(x_0)(x - x_0).$$

Geometrically, the secant line approaches a unique limiting line. If that line is not vertical and its slope is finite, the limiting slope is the derivative.



§1.2 Derivative notation and local definition

The definition requires f to be defined on a punctured neighborhood of x_0 , and usually on a small open interval around it. The derivative at x_0 may be denoted by any of

$$f'(x_0), \quad \left. \frac{df}{dx} \right|_{x=x_0}, \quad y'(x_0), \quad \left. \frac{dy}{dx} \right|_{x=x_0}.$$

If f is differentiable at every point of an open interval (a, b) , then $x \mapsto f'(x)$ is the derivative function on that interval.

Before asking whether f is differentiable at a point, one must know that it is defined on a neighborhood of that point. Differentiability is a local property.

§1.3 Rules of differentiation

If f and g are differentiable at x_0 , then sums, scalar multiples, products, and quotients obey:

$$(af + bg)' = af' + bg', \quad (fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \quad g(x_0) \neq 0.$$

The chain rule says

$$(f \circ g)'(x_0) = f'(g(x_0))g'(x_0).$$

The inverse-function rule says that if f and f^{-1} are inverse functions, $f'(x_0) \neq 0$, and $y_0 = f(x_0)$, then

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

These rules are not simply a table to memorize. Each rule is a disguised statement about first-order linear approximation. For example,

$$f(x_0 + h) = f(x_0) + f'(x_0)h + o(h), \quad g(x_0 + h) = g(x_0) + g'(x_0)h + o(h),$$

and multiplying these expansions gives the product rule.

§1.4 Elementary derivative formulas

The table in the note includes the standard formulas:

$$C' = 0, \quad (x^a)' = ax^{a-1}, \quad (a^x)' = a^x \ln a, \quad (\log_a x)' = \frac{1}{x \ln a}.$$

For trigonometric functions,

$$\begin{aligned} (\sin x)' &= \cos x, & (\cos x)' &= -\sin x, & (\tan x)' &= \sec^2 x, & (\cot x)' &= -\csc^2 x, \\ (\sec x)' &= \sec x \tan x, & (\csc x)' &= -\csc x \cot x. \end{aligned}$$

For inverse trigonometric functions,

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, \quad (\arccos x)' = -\frac{1}{\sqrt{1-x^2}}, \quad (\arctan x)' = \frac{1}{1+x^2}.$$

§1.5 Extrema, Fermat’s lemma, and critical points

A point x_0 is a local maximum if $f(x) \leq f(x_0)$ near x_0 , and a local minimum if $f(x) \geq f(x_0)$ near x_0 . If $f'(x_0) = 0$, then x_0 is called a stationary or critical point.

Fermat’s lemma says: if f is differentiable at x_0 and has a local extremum there, then

$$f'(x_0) = 0.$$

The note writes a warning beside this theorem: the converse is false. For example,

$$f(x) = x^3$$

has $f'(0) = 0$, but 0 is not an extremum. Fermat’s lemma only provides candidates for extrema. One must then check sign changes, second derivatives, endpoints, or non-differentiable points.

If $f''(x_0) > 0$ and $f'(x_0) = 0$, then x_0 is a local minimum. If $f''(x_0) < 0$, it is a local maximum. The reason is that f'' describes whether f' is increasing or decreasing near the critical point.

§1.6 A derivative proof of a basic inequality

For $x > 0$, one has

$$x^\alpha - \alpha x + \alpha - 1 \leq 0, \quad 0 < \alpha < 1,$$

and

$$x^\alpha - \alpha x + \alpha - 1 \geq 0, \quad \alpha < 0 \text{ or } \alpha > 1.$$

Let

$$F(x) = x^\alpha - \alpha x + \alpha - 1.$$

Then

$$F'(x) = \alpha(x^{\alpha-1} - 1), \quad F(1) = 0.$$

For $0 < \alpha < 1$, F increases on $(0, 1)$ and decreases on $(1, \infty)$, so $x = 1$ is a maximum and $F(x) \leq 0$. For $\alpha < 0$ or $\alpha > 1$, the monotonicity is reversed, and $x = 1$ is a minimum, so $F(x) \geq 0$.

This small inequality is the engine behind Young’s inequality.

§1.7 Young’s inequality

Let $a, b > 0$ and let p, q satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

For $p > 1$, Young’s inequality says

$$a^{1/p} b^{1/q} \leq \frac{a}{p} + \frac{b}{q},$$

with equality iff $a = b$. The proof substitutes

$$x = \frac{a}{b}, \quad \alpha = \frac{1}{p}$$

into the previous derivative inequality, then multiplies by b .

The source also records the reversed version for $p < 1$, where the inequality sign changes. The important conceptual point is that an inequality between weighted geometric and arithmetic means is reduced to the shape of one single-variable function.

§1.8 Holder's inequality

For $x_i, y_i \geq 0$ and $1/p + 1/q = 1$, Holder's inequality is

$$\sum_{i=1}^n x_i y_i \leq \left(\sum_{i=1}^n x_i^p \right)^{1/p} \left(\sum_{i=1}^n y_i^q \right)^{1/q}, \quad p > 1.$$

The proof in the source normalizes

$$a_i = \frac{x_i^p}{\sum_j x_j^p}, \quad b_i = \frac{y_i^q}{\sum_j y_j^q}$$

and applies Young's inequality termwise. After adding the inequalities, the normalized sum becomes 1, giving Holder.

The integral form follows the same idea:

$$\int_a^b f(x)g(x) dx \leq \left(\int_a^b |f(x)|^p dx \right)^{1/p} \left(\int_a^b |g(x)|^q dx \right)^{1/q}.$$

The comment says this feels almost trivial once the finite-sum proof is understood: one replaces sums by integrals and uses the same normalization logic.

For $p = q = 2$, Holder becomes the Cauchy–Bunyakovsky–Schwarz inequality.

§1.9 Minkowski's inequality

For $p > 1$, Minkowski's inequality says

$$\left(\sum_{i=1}^n (x_i + y_i)^p \right)^{1/p} \leq \left(\sum_{i=1}^n x_i^p \right)^{1/p} + \left(\sum_{i=1}^n y_i^p \right)^{1/p}.$$

The proof applies Holder to the two sums

$$\sum_i x_i (x_i + y_i)^{p-1}, \quad \sum_i y_i (x_i + y_i)^{p-1}.$$

Because $q = p/(p-1)$, we have

$$((x_i + y_i)^{p-1})^q = (x_i + y_i)^p.$$

After factoring the common term and dividing by it, one obtains Minkowski.

For $p = 2$, this is exactly the triangle inequality for Euclidean length:

$$\sqrt{(x_1 + y_1)^2 + \cdots + (x_n + y_n)^2} \leq \sqrt{x_1^2 + \cdots + x_n^2} + \sqrt{y_1^2 + \cdots + y_n^2}.$$

§1.10 Convex functions and Jensen's inequality

The note defines a convex function by the inequality

$$f(\alpha_1 x_1 + \alpha_2 x_2) \leq \alpha_1 f(x_1) + \alpha_2 f(x_2), \quad \alpha_1 + \alpha_2 = 1, \quad \alpha_i \geq 0.$$

Geometrically, the point on the chord between $(x_1, f(x_1))$ and $(x_2, f(x_2))$ lies above the graph.

Equivalently, for $x_1 < x < x_2$,

$$\frac{f(x) - f(x_1)}{x - x_1} \leq \frac{f(x_2) - f(x)}{x_2 - x}.$$

This slope inequality implies that if f is differentiable, then f' is increasing; if f is twice differentiable, then convexity corresponds to $f'' \geq 0$.

Jensen's inequality generalizes the two-point definition:

$$f\left(\sum_{i=1}^n \alpha_i x_i\right) \leq \sum_{i=1}^n \alpha_i f(x_i), \quad \alpha_i \geq 0, \quad \sum_i \alpha_i = 1.$$

The source proves this by induction. Let $\beta = \alpha_2 + \cdots + \alpha_n = 1 - \alpha_1$ and normalize the remaining weights by α_i/β . Then use the two-point convexity inequality followed by the induction hypothesis.

§1.11 Typical exercise themes

The last pages list exercises. They are not random; they are organized around the derivative as a tool for inequality and extrema.

Examples include:

- (i) determining when a line and cubic have three distinct intersections;
- (ii) finding monotonicity intervals of $f(x) = \sqrt{x} - \ln(x + a)$;
- (iii) maximizing $\sin^2 x \cos x$ on $(0, \pi/2)$;
- (iv) comparing numbers such as $0.1e^{0.1}$, $1/9$, and $-\ln 0.9$ using auxiliary functions;
- (v) proving $e^x \geq 1 + x$ and $\ln x \leq x - 1$;
- (vi) using Jensen or logarithms to prove entropy-like inequalities such as

$$\sum_{i=1}^{2^n} p_i \log_2 p_i \geq -n, \quad \sum_i p_i = 1;$$

- (vii) proving product inequalities of the form

$$\prod_{i=1}^n a_i^{b_i} \leq 1$$

under weighted-sum hypotheses;

- (viii) minimizing expressions such as

$$\frac{9}{\sin^2 x} + \frac{1}{\cos x \cos y \sin^2 y}.$$

The common method is to build an auxiliary function, compute derivatives, locate monotonicity or convexity, and then read off the desired inequality.

§1.12 The reusable pattern

This note shows how the derivative becomes more than a tangent slope. It gives a language for extrema, monotonicity, convexity, and inequalities. Young, Holder, Minkowski, and Jensen all look like different inequalities, but the note reveals a chain:

$$\begin{aligned} \text{single-variable derivative inequality} &\rightarrow \text{Young} \rightarrow \text{Holder} \\ &\rightarrow \text{Minkowski}, \quad \text{convexity} \rightarrow \text{Jensen}. \end{aligned}$$

Thus calculus becomes a proof technology for analysis and inequalities.

§1.13 Derivative as local linear approximation

The derivative is the coefficient of the best first-order model:

$$f(a+h) = f(a) + f'(a)h + o(h).$$

This viewpoint explains the product, quotient, and chain rules: they are rules for composing and multiplying first-order approximations.

In several variables the same idea becomes the Fréchet derivative, a linear map rather than a single number.

§1.14 Extrema and first variation

Fermat's lemma says that if f has a local extremum at an interior point and is differentiable there, then $f'(a) = 0$. The derivative is the first variation; if a small positive and negative move are both allowed, the linear term must vanish.

Second derivative tests then study the next nonzero term in the local expansion.

§1.15 Young, Holder, and convex duality

Young's inequality

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}, \quad \frac{1}{p} + \frac{1}{q} = 1,$$

is a convex-duality statement. It is the pointwise engine behind Holder's inequality. Minkowski then gives the triangle inequality for L^p norms.

Thus the inequality part of the note connects differential calculus to convex analysis and normed spaces.

§1.16 Local linearity and convex control

The chapter links two central ideas: derivatives provide local linear models, and inequalities describe convex or norm geometry. Tangents, extrema, Young–Holder–Minkowski, and Jensen all express how local or averaged information controls global behavior.